OR4023 (Optimization) Homework 3

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Question 1. Let $f(x) = x^4 - 2x^3$.

(1) Find the global minimum of f over $[-1, 2]$ in the following steps.

   (a) Apply the first order necessary conditions for interior points to find candidates for local minimum points.

   (b) Apply the first order necessary conditions for the boundary points to find candidate for local minimum points.

   (c) Compare function values at the points in (a) and (b) to find the global minimum points.

(2) Apply the second order sufficient conditions to determine which points found in (1a) and (1b) are (strict) local minimum points.

Solution.

(1) (a) Compute the first derivative:

$$f'(x) = 4x^3 - 6x^2 = 2x^2(2x - 3).$$

The interior critical points are $x = 0$ and $x = \frac{3}{2}$.

   (b) The boundary points are $x = -1$ and $x = 2$. Since $f'(-1) = -10$, $x = -1$ is not a local minimizer. Since $f'(2) = 8 > 0$, $x = 2$ is not a local minimizer.

   (c) Evaluating $f(x)$ at cirtical points, we have:

$$f(0) = 0, \quad f\left(\frac{3}{2}\right) = -\frac{27}{16}.$$

Hence, the global minimum occurs at $x = \frac{3}{2}$ with $f\left(\frac{3}{2}\right) = -\frac{27}{16}$.

(2) The second derivative is

$$f''(x) = 12x^2 - 12x = 12x(x - 1).$$

At $x = \frac{3}{2}$, $f''\left(\frac{3}{2}\right) > 0$, confirming a strict local minimum at $x = \frac{3}{2}$. At $x = 0$, $f''(0) = 0$, the second order sufficient condition does not apply. But it is actually not a local minizer (check by yourself by definition).

Therefore, the global minimum of f over $[-1, 2]$ is a strict local minimum at $x = \frac{3}{2}$.

Question 2. Find a strict local minimum of $f(x, y) = x^4 + 2y^2 - xy$ over $\mathbb{R}^2$ using the second order sufficient conditions.

Solution.

Step 1: Compute the gradient of f and set it to zero.

$$\nabla f(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} 4x^3 - y \\ 4y - x \end{pmatrix}$$

Setting $\nabla f(x, y) = 0$, we get:

$$4x^3 - y = 0$$
$$4y - x = 0$$

From the second equation, we have $y = \frac{x}{4}$. Substituting this into the first equation:

$$4x^3 - \frac{x}{4} = 0$$

This gives us $x = 0$ or $x = \pm\frac{1}{4}$.
When $x = 0$, we have $y = 0$.
When $x = \frac{1}{4}$, we have $y = \frac{1}{16}$.
When $x = -\frac{1}{4}$, we have $y = -\frac{1}{16}$.
So, we have three critical points: $(0, 0)$, $(\frac{1}{4}, \frac{1}{16})$, and $(-\frac{1}{4}, -\frac{1}{16})$.

Step 2: Apply the second order sufficient conditions.
The Hessian matrix of f is:

$$H(x, y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} 12x^2 & -1 \\ -1 & 4 \end{pmatrix}$$

For a critical point to be a strict local minimum, the Hessian matrix must be positive definite at that point.

At $(0, 0)$:

$$H(0, 0) = \begin{pmatrix} 0 & -1 \\ -1 & 4 \end{pmatrix}$$

The determinant of $H(0, 0)$ is $0 \cdot 4 - (-1) \cdot (-1) = 0 - 1 = -1 < 0$, so $H(0, 0)$ is not positive definite. No conclusion can be made about $(0, 0)$ from the second order sufficient conditions. But it is actually not a local minizer due to second order necessary conditions (check by yourself using second order necessary conditions).

At $(\frac{1}{4}, \frac{1}{16})$ and $(-\frac{1}{4}, -\frac{1}{16})$:

$$H = \begin{pmatrix} \frac{3}{4} & -1 \\ -1 & 4 \end{pmatrix}$$

The determinant of H is $\frac{3}{4} \cdot 4 - (-1) \cdot (-1) = 3 - 1 = 2 > 0$, and the first leading principal minor is $\frac{3}{4} > 0$. Therefore, $H\left(\frac{1}{4}, \frac{1}{16}\right)$ is positive definite, and $\left(\frac{1}{4}, \frac{1}{16}\right)$ and $(-\frac{1}{4}, -\frac{1}{16})$ are both strict local minimizers.

Conclusion: The function $f(x, y) = x^4 + 2y^2 - xy$ has two strict local minima at $\left(\frac{1}{4}, \frac{1}{16}\right)$ and $\left(-\frac{1}{4}, -\frac{1}{16}\right)$.

Question 3. Let $b \in \mathbb{R}^n$ be a vector and $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix.

(a) If $f(x) = b^T x$, show $\nabla f(x) = b$ and $\nabla^2 f(x) = 0$.

(b) If $f(x) = \frac{1}{2} x^T A x$, show $\nabla f(x) = Ax$ and $\nabla^2 f(x) = A$.

Solution.

(a) For $f(x) = b^T x$, write

$$f(x) = \sum_{i=1}^{n} b_i x_i.$$

Differentiating with respect to x_j, we find

$$\frac{\partial f}{\partial x_j} = b_j.$$

Hence, the gradient is

$$\nabla f(x) = (b_1, b_2, \ldots, b_n)^T,$$

Also, since the first derivatives are constants, the Hessian (matrix of second derivatives) is

$$\nabla^2 f(x) = 0.$$

(b) For $f(x) = \frac{1}{2}x^T Ax$, first note that

$$f(x) = \frac{1}{2}\sum_{i,j=1}^{n} x_i A_{ij} x_j.$$

Differentiating with respect to x_k, and using the symmetry of A,

$$\frac{\partial f}{\partial x_k} = \frac{1}{2}\sum_{i=1}^{n} A_{ik}x_i + \frac{1}{2}\sum_{j=1}^{n} A_{kj}x_j = \sum_{j=1}^{n} A_{kj}x_j.$$

Therefore, the gradient is

$$\nabla f(x) = Ax,$$

Differentiating again, the Hessian is the constant matrix

$$\nabla^2 f(x) = A.$$

Question 4. Let

$$f(x_1, x_2) = x_1^2 + 4x_2^2 - 6x_1x_2 + 5x_1 - 7x_2 + 8.$$

Let $x = (x_1, x_2) \in \mathbb{R}^2$. Find a symmetric matrix A, a vector b and a number c so that

$$f(x) = \frac{1}{2}x^T Ax + b^T x + c.$$

Solution. To express $f(x_1, x_2) = x_1^2 + 4x_2^2 - 6x_1x_2 + 5x_1 - 7x_2 + 8$ in the form $\frac{1}{2}x^T Ax + b^T x + c$, we need to identify the quadratic, linear, and constant terms.

First, let's identify the quadratic terms to determine matrix A. For a quadratic form $\frac{1}{2}x^T Ax$, we have:

$$\frac{1}{2}x^T Ax = \frac{1}{2}\begin{pmatrix} x_1 & x_2 \end{pmatrix}\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Expanding this expression:

$$\frac{1}{2}x^T Ax = \frac{1}{2}(a_{11}x_1^2 + a_{12}x_1x_2 + a_{21}x_2x_1 + a_{22}x_2^2)$$

Since A is symmetric, $a_{12} = a_{21}$, so:

$$\frac{1}{2}x^T Ax = \frac{1}{2}(a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2)$$

Comparing with the quadratic terms in $f(x_1, x_2)$, we have $a_{11} = 2$, $a_{22} = 8$, and $a_{12} = -6$. Therefore, the symmetric matrix A is:

$$A = \begin{pmatrix} 2 & -6 \\ -6 & 8 \end{pmatrix}$$

For the linear term $b^T x$, we have:

$$b^T x = \begin{pmatrix} b_1 & b_2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = b_1 x_1 + b_2 x_2$$

Comparing with the linear terms in $f(x_1, x_2)$, we obtain $b_1 = 5$ and $b_2 = -7$. Therefore, the vector b is:

$$b = \begin{pmatrix} 5 \\ -7 \end{pmatrix}$$

Finally, the constant term $c = 8$.

In conclusion, the function $f(x_1, x_2) = x_1^2 + 4x_2^2 - 6x_1x_2 + 5x_1 - 7x_2 + 8$ can be expressed as:

$$f(x) = \frac{1}{2} x^T A x + b^T x + c$$

where

$$A = \begin{pmatrix} 2 & -6 \\ -6 & 8 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ -7 \end{pmatrix}, \quad c = 8$$

Question 5. Let $f(x) = x^2 + e^x, x \in [-1, 1]$.

(a) Show that f has a unique minimizer, which is strict and global.

(b) Start from the initial point $x_0 = 0$, apply 2 iterations of the Newton's method to find x_2.

(c) Start from the initial points $x_0 = -1, x_1 = 1$, apply 2 iterations of the secant method to find x_3.

Solution.

(a) To show that $f(x) = x^2 + e^x$ has a unique minimizer on $[-1, 1]$, we'll find all critical points and analyze them.

First, we compute the first derivative:

$$f'(x) = 2x + e^x$$

Let $g(x) = 2x + e^x$. Note that $g'(x) = 2 + e^x > 0$ for all x, so $g(x)$ is strictly increasing.

Evaluating at the endpoints:

$$g(-1) = -2 + e^{-1} < 0, \quad g(1) = 2 + e > 0.$$

By the Intermediate Value Theorem, there exists exactly one value $x^* \in (-1, 1)$ such that $g(x^*) = 0$. This is our only critical point.

To verify it's a strict local minimum and also the global minimum, we note that:

$$f''(x) = 2 + e^x > 0 \quad \text{for all } x$$

Since $f''(x) > 0$ for all $x \in [-1, 1]$, the function f is strictly convex on $[-1, 1]$. For a convex function, any local minimum is automatically a global minimum. Therefore, the unique critical point x^* is both a strict local minimum and the global minimum on $[-1, 1]$.

(b) For Newton's method, we use the iteration formula:

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}.$$

For our function $f(x) = x^2 + e^x$:

$$f'(x) = 2x + e^x, \quad f''(x) = 2 + e^x.$$

Starting with $x_0 = 0$:

First iteration:

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)} = -\frac{1}{3}.$$

Second iteration:

$$x_2 = x_1 - \frac{f'(x_1)}{f''(x_1)} \approx -0.3517.$$

(c) **First iteration:**

$$x_2 = x_1 - \frac{f'(x_1)(x_1 - x_0)}{f'(x_1) - f'(x_0)} = \frac{-e - e^{-1}}{4 + e - e^{-1}} \approx -0.486$$

Second iteration:

We first compute $f'(x_2)$:

$$f'(x_2) = 2x_2 + e^{x_2} \approx -0.357$$

Now we can compute x_3:

$$x_3 = x_2 - \frac{f'(x_2)(x_2 - x_1)}{f'(x_2) - f'(x_1)} \approx -0.382$$

Question 6. For each of the following sequences, determine their limit and the order of convergence (or superlinear if there is no order).

(a) $r_k = \frac{1}{k} - \frac{1}{k+1}, k \geqslant 1$

(b) $r_{k+1} = \frac{r_k}{k+1}, k \geqslant 1$

Solution. (a) Consider the sequence $r_k = \frac{1}{k} - \frac{1}{k+1}$.

Limit:

$$r_k = \frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)}$$

As $k \to \infty$, $\frac{1}{k(k+1)} \to 0$. Therefore, the limit of r_k is 0.

Order of Convergence: To determine the order of convergence, we need to examine $\lim_{k\to\infty} \frac{|r_{k+1}|}{|r_k|^p}$ for some $p > 0$. For our sequence $r_k = \frac{1}{k(k+1)}$:

$$\frac{|r_{k+1}|}{|r_k|^p} = \frac{\frac{1}{(k+1)(k+2)}}{\left(\frac{1}{k(k+1)}\right)^p} = \frac{k^p(k+1)^p}{(k+1)(k+2)} = \frac{k^p(k+1)^{p-1}}{k+2}$$

When $p = 1$:

$$\lim_{k\to\infty} \frac{|r_{k+1}|}{|r_k|} = \lim_{k\to\infty} \frac{k}{k+2} = 1$$

Therefore, the sequence converges linearly (order 1) to 0.

(b) Consider the sequence $r_{k+1} = \frac{r_k}{k+1}$.

Limit:

$$r_2 = \frac{r_1}{2}$$
$$r_3 = \frac{r_2}{3} = \frac{r_1}{2 \cdot 3}$$
$$r_4 = \frac{r_3}{4} = \frac{r_1}{2 \cdot 3 \cdot 4}$$

By induction, we can show that $r_k = \frac{r_1}{k!}$ for $k \geq 1$. Since $\lim_{k\to\infty} \frac{1}{k!} = 0$ for any fixed r_1, we have $\lim_{k\to\infty} r_k = 0$. Therefore, the limit of r_k is 0.

Order of Convergence: Since

$$\frac{|r_{k+1}|}{|r_k|} = \frac{1}{k+1} \to 0 \text{ as } k \to \infty,$$

it follows by definition that r_k converges to 0 superlinearly. (Question: Can you prove the convergence order can not be higher or less than 1?)